25900112 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCHTC100 - PLANT PROPAGATION

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **1** marks

1. Define vegetative mode of reproduction. [K] / [CO1]
2. Write any two importance of artificial method of plant propagation. [U] / [CO2]
3. Which vitamin is essential for coenzyme formation in plant tissue culture? [K] / [CO2]
4. How does micropropagation differ from traditional plant propagation methods. [U] / [CO1]
5. What is Wedge Grafting? [K] / [CO3]
6. Mention the purpose of mist chamber in plant propagation. [U] / [CO4]
7. Write an example for plant that propagated through stem cutting. [K] / [CO3]
8. Define micropropagation. [U] / [CO3]
9. Name the propagules that are propagated by division. [K] / [CO2]
10. Identify three essential utilities needed for a functional micropropagation laboratory. [K] / [CO2]

11. What are offsets? [K] / [CO3]

12. Why is layering used? [U] / [CO3]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Why is gelling agent concentration important in micropropagation media? [U] / [CO2]

14. How does the nutritional state of parent plant impact rooting of cuttings? [U] / [CO3]

15. Differentiate between dry heat and moist heat sterilization methods. [U] / [CO2]

16. Explain the uses of different tools in plant propagation. [K] / [CO1]

17. What are the key features of phytotrons? [U] / [CO4]

18. Explain bulbs, pseudobulbs, corms and tubers with examples. [U] / [CO3]

19. How is irrigation managed in a polyhouse, and what systems are most commonly used? [U] / [CO4]

20. How do auxins influence rooting in plants? [U] / [CO3]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Describe the unorganized methods of plant micropropagation and explain their significance in plant tissue culture. [U] / [CO3]

22. Describe the factors influencing rooting in cuttings. [U] / [CO3]

[10x1 = 10]